AI / LLM USAGE LOG

Coding assistant used: GitHub Copilot (VS Code) — an LLM-based coding assistant.

#	Date	Task	Prompt / question	What was used	Student verification
1	2026-11-02	Socket API basics	“Explain the sequence socket → bind → listen → accept → recv/send → close and what each call returns on error.”	Explanation only	Re-read man 2 socket and man 2 accept; confirmed return values and errno handling in server.c
2	2026-11-02	Concurrency choice	“Compare thread-per-client, select(), poll() and fork() for a small TCP server. Which is easiest to explain in a viva?”	Concept / recommendation	Chose pthreads; wrote the rationale in REPORT §10.2; verified 3 simultaneous clients in the test suite
3	2026-11-02	Protocol design	“Design a line-oriented text protocol for email with subjects and bodies that may contain | and newlines.”	Design suggestion (escaping scheme)	Implemented proto_escape/proto_unescape; added the “escaped body round-trips” test
4	2026-11-02	Framing implementation	“Write a recv_line() that reads until newline, strips carriage return, and reports overflow without desynchronising the stream.”	Code suggestion	Reviewed the drain-on-overflow logic; tested with a 20,000-byte line and confirmed the connection still answers INBOX
5	2026-11-02	Password storage	“How should I store passwords in a file-based C app without a crypto library?”	Concept + SHA-256 implementation sketch	Implemented sha256.c and verified it against sha256sum. Later removed on request: the assignment does not require hashing, so the final version stores passwords as plain text to keep the code simple. The trade-off is documented in REPORT §15/§16
6	2026-11-02	Storage layout	“Suggest a simple file layout for users and per-user inbox/sent boxes.”	Design suggestion	Adopted users.dat + mail/<user>/{inbox,sent}.dat; inspected files with cat after a live session
7	2026-11-02	Atomic mailbox update	“How do I mark one record as read in a text file safely?”	Concept (write temp + rename)	Implemented rewrite_box(); verified read/unread transitions and deletion in tests
8	2026-11-02	Compiler errors	“Fix implicit declaration of strcasecmp and va_start under -std=c11.”	Code fix	Added <strings.h> and <stdarg.h>; rebuild is warning-free
9	2026-11-02	Buffer warnings	“Fix -Wformat-truncation warnings on snprintf into fixed buffers.”	Code fix	Bounded the body copy with %.*s; rebuild is warning-free
10	2026-11-02	Test matrix	“List test cases for registration, login, send, inbox, read, sent, delete, malformed input and concurrency.”	Test case list	Implemented as tests/test_protocol.py; all 44 checks pass
11	2026-11-02	Bug found by tests	“The escaped body round-trips test fails: a subject containing | is split into extra fields.”	Diagnosis + fix (proto_unescape_inplace applied to payload fields in handle_line)	Confirmed the root cause (client escaped, server never unescaped); applied the fix; test now passes; re-ran the full suite
12	2026-11-02	Documentation	“Draft a README with compilation, execution, port, configuration, data locations, examples and limitations.”	Draft text	Rewrote in own words; verified every command by running it
13	2026-11-02	Report structure	“Suggest a structure for the project report matching the assignment’s recommended sections.”	Outline	Followed the assignment’s 19-section structure; wrote the content from the actual implementation
14	2026-11-02	Error handling review	“Review the server for paths where a bad client request could crash it.”	Review comments	Added SIGPIPE ignore, per-connection error paths, pthread_create failure handling; tested with binary garbage and abrupt disconnects